



Recitation 14

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1. Homework

2. Review

3. Example

4. Recitation



1. Homework

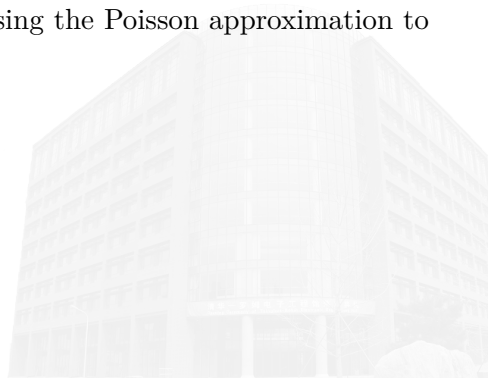
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1. During each day, the probability that your computer's operating system crashes at least once is 5%, independent of every other day. You are interested in the probability of at least 45 crash-free days out of the next 50 days.
- (a) Find the probability of interest by using the normal approximation to the binomial.
 - (b) Repeat part (a), this time using the Poisson approximation to the binomial.



Note 1: There is a typo in the problem set when copying the problem from the book. “You are interested in the 45 crash-free days out of the next 50 days.” Therefore, calculating either $\mathbf{P}(S \geq 45)$ or $\mathbf{P}(S = 45)$ is correct.

Note 2: It is better using $1/2$ correction when estimating probability of discrete RVs.

Note 3: Poisson approximation is more accurate for large n and small p . Therefore, we should use $p' = 0.05$ and approximate $\mathbf{P}(S' \leq 5)$ or $\mathbf{P}(S' = 5)$.

4. It is well known that infants born to mothers who smoke tend to be small and prone to a range of ailments. It is conjectured that also they look abnormal. Nurses were shown selections of photographs of babies, one half of whom had smokers as mothers; the nurses were asked to judge from a baby's appearance whether or not the mother smoked. In 1500 trials the correct answer was given 910 times. Is the conjecture plausible? If so, why?



If the conjecture is not plausible, it means that the nurses judge the babies by pure guessing, i.e., the probability of correct judgment is $p = 0.5$.

Then, we apply CLT to show that if the conjecture is not plausible, the probability of 910 correct answers out of 1500 samples is hardly likely.

$$\begin{aligned}\mathbf{P}(X \geq 910) &= \mathbf{P}\left(\frac{X - \frac{1}{2}n}{\frac{1}{2}\sqrt{n}} \geq \frac{910 - 750}{5\sqrt{15}}\right) \\ &\approx 1 - \Phi(8.26) \approx 0\end{aligned}$$

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- Bernoulli process

- A sequence of independent Bernoulli RVs with common parameter p
- Independence and memorylessness
- Interarrival times: Geometric RVs with common parameter p
- The k th arrival time: Pascal PMF of order k

$$p_{Y_k}(t) = \binom{t-1}{k-1} p^k (1-p)^{t-k}, \quad t = k, k+1, \dots$$

$$\mathbf{E}[Y_k] = \mathbf{E}[T_1] + \dots + \mathbf{E}[T_k] = \frac{k}{p}$$

$$\text{var}(Y_k) = \text{var}(T_1) + \dots + \text{var}(T_k) = \frac{k(1-p)}{p^2}$$

- Splitting: $\text{Bernoulli}(p) \rightarrow \text{Bernoulli}(pq) \ \& \ \text{Bernoulli}(p(1-q))$
- Merging: $\text{Bernoulli}(p) \ \& \ \text{Bernoulli}(q) \rightarrow \text{Bernoulli}(p+q-pq)$

- Poisson process

- Time homogeneity & Independence & Small interval probabilities
- PMF of number of arrivals:

$$\mathbf{P}(k, \tau) = \frac{(\lambda\tau)^k e^{-\lambda\tau}}{k!}, \quad k = 0, 1, \dots$$

$$\mathbf{E}[N_t] = \lambda t, \text{ var}(N_t) = \lambda t$$

- Independence and memorylessness
- Interarrival times: Exponential RVs with common parameter λ
- The k th arrival time: Erlang distribution

$$f_{Y_k}(y) = \frac{\lambda^k y^{k-1} e^{-\lambda y}}{(k-1)!}, \quad y \geq 0$$

- Merging: $\text{Poisson}(\lambda) \& \text{Poisson}(\mu) \rightarrow \text{Poisson}(\lambda + \mu)$

$$\mathbf{P}(\text{The first process arrives} \mid 1 \text{ arrival}) \approx \frac{\lambda}{\lambda + \mu}$$

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- You are told that you are the 56th person in line. Callers depart according to a Poisson process with a rate of $\lambda = 2$ per minute. How long will you have to wait on the average, and what is the probability you will wait for more than 30 minutes?



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- The waiting time in minutes, defined as Y , is Erlang of order 56, and

$$\mathbf{E}[Y] = \frac{56}{\lambda} = 28.$$

- The probability that you have to wait for more than 30 minutes is given by

$$\mathbf{P}(Y \geq 30) = \int_{30}^{\infty} \frac{\lambda^{56} y^{55} e^{-\lambda y}}{55!} dy.$$

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1. Let X , Y , and Z be independent exponential RVs with parameters λ , μ , and ν , respectively. Find $\mathbf{P}(X < Y < Z)$.



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Solution:

The event $\{X < Y < Z\}$ can be expressed as

$$\{X < \min\{Y, Z\}\} \cap \{Y < Z\}$$

where

$$\mathbf{P}(Y < Z) = \frac{\mu}{\mu + \nu},$$

$$\mathbf{P}(X < \min\{Y, Z\}) = \frac{\lambda}{\lambda + \mu + \nu}.$$

Note that the event $\{X < \min\{Y, Z\}\}$ and $\{Y < Z\}$ are independent. Hence, we have

$$\mathbf{P}(X < Y < Z) = \frac{\lambda\mu}{(\lambda + \mu + \nu)(\mu + \nu)}.$$

2. Consider a Poisson process with parameter λ , and an independent random variable T , which is exponential with parameter μ . Find the PMF of the number of Poisson arrivals during the time interval $[0, T]$.



Let us view T as the first arrival time of a new, independent Poisson process with parameter μ , and merge the process with the original Poisson process. Therefore, each arrival in the merged process comes from the original Poisson process with probability $\lambda/(\lambda + \mu)$, independent of other arrivals.

If we view an arrival from the new process as a success, we note that the number K of arrivals until the first success has a geometric PMF with parameter $\mu/(\lambda + \mu)$, i.e.,

$$p_K(k) = \left(\frac{\mu}{\lambda + \mu} \right) \left(\frac{\lambda}{\lambda + \mu} \right)^{k-1}, \quad k = 1, 2, \dots$$

The number of arrivals from the original Poisson process until the first success is equal to $K - 1$, so the PMF we want is

$$p_N(n) = p_K(n + 1) = \left(\frac{\mu}{\lambda + \mu} \right) \left(\frac{\lambda}{\lambda + \mu} \right)^n, \quad n = 0, 1, \dots$$